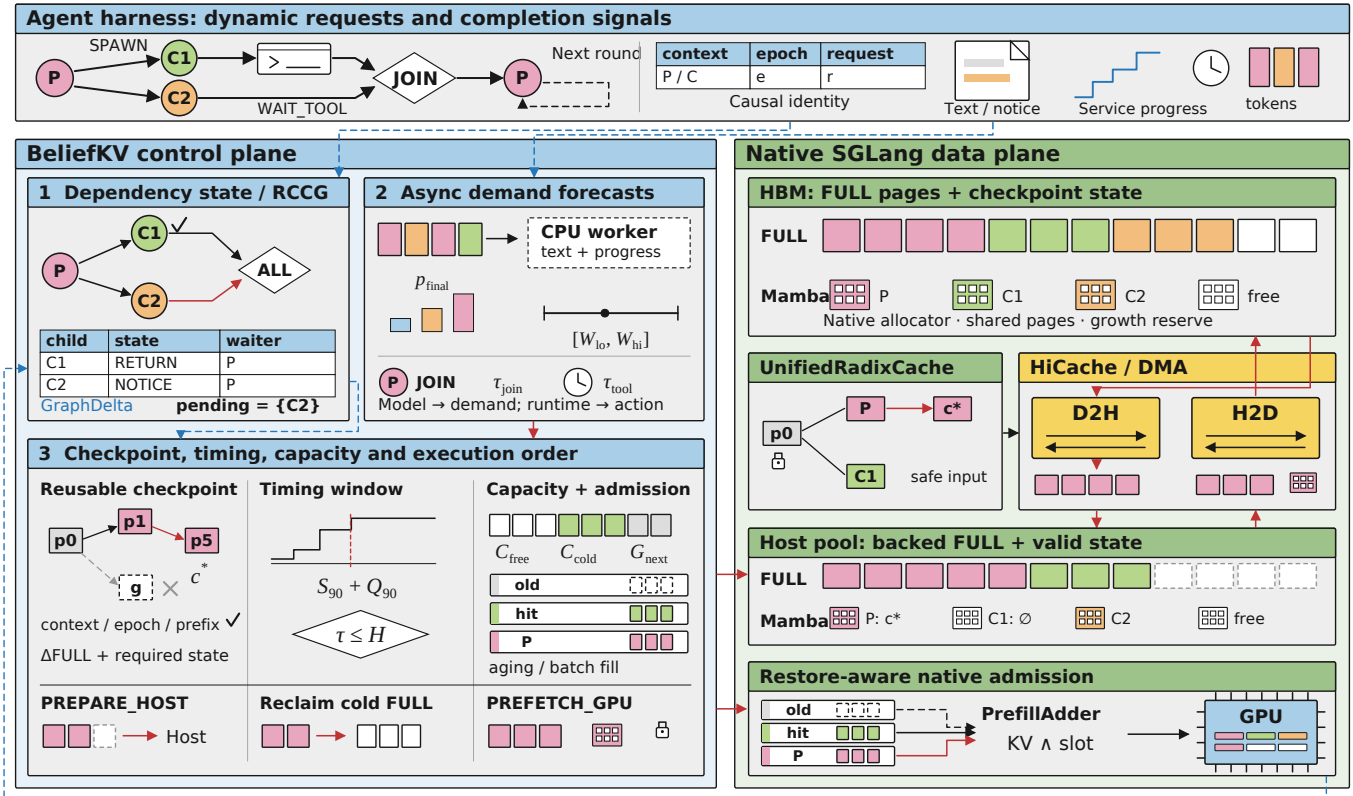
BeliefKV: Workflow-aware Execution and KV Management

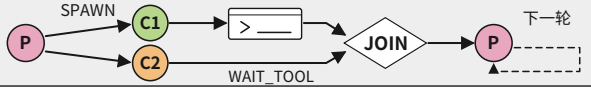
Observed dependencies → demand forecasts → page migration and native admission



BeliefKV：工作流感知的执行与 KV 协同管理

运行时依赖 → 近期执行需求 → 页级迁移、驻留与原生准入

Agent harness：动态请求与完成信号



context	epoch	request
P / C	e	r

因果身份

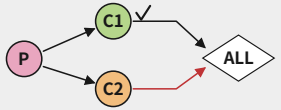
正文 / 通知

服务进度

tokens

BeliefKV 控制面

1 依赖状态 / RCCG



child	state	waiter
C1	RETURN	P
C2	NOTICE	P

GraphDelta　　pending = {C2}

2 异步执行需求预测



p_{final}

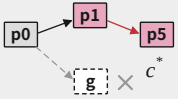
$[W_{\text{lo}}, W_{\text{hi}}]$

P JOIN　τ_{join}　τ_{tool}

模型预测需求；runtime 决定动作

3 Checkpoint、时机、容量与执行顺序

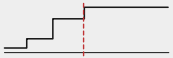
可复用 checkpoint



context / epoch / prefix ✓

ΔFULL + required state

实测传输启动窗口



$S_{90} + Q_{90}$

$\tau \le H$

容量与准入机会



aging / batch fill

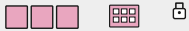
PREPARE_HOST



回收冷 FULL

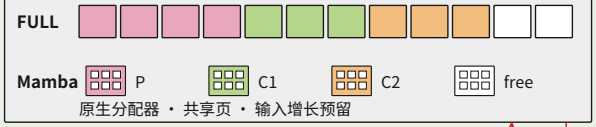


PREFETCH_GPU

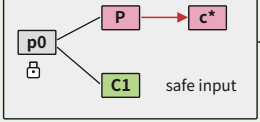


原生 SGLang 数据面

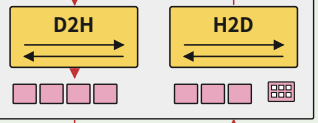
HBM: FULL pages + checkpoint state



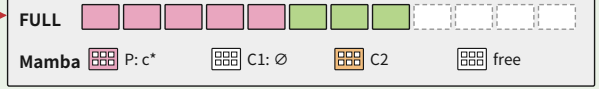
UnifiedRadixCache



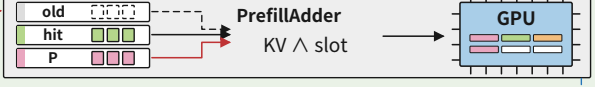
HiCache / DMA



Host 池：已备份 FULL 与有效状态



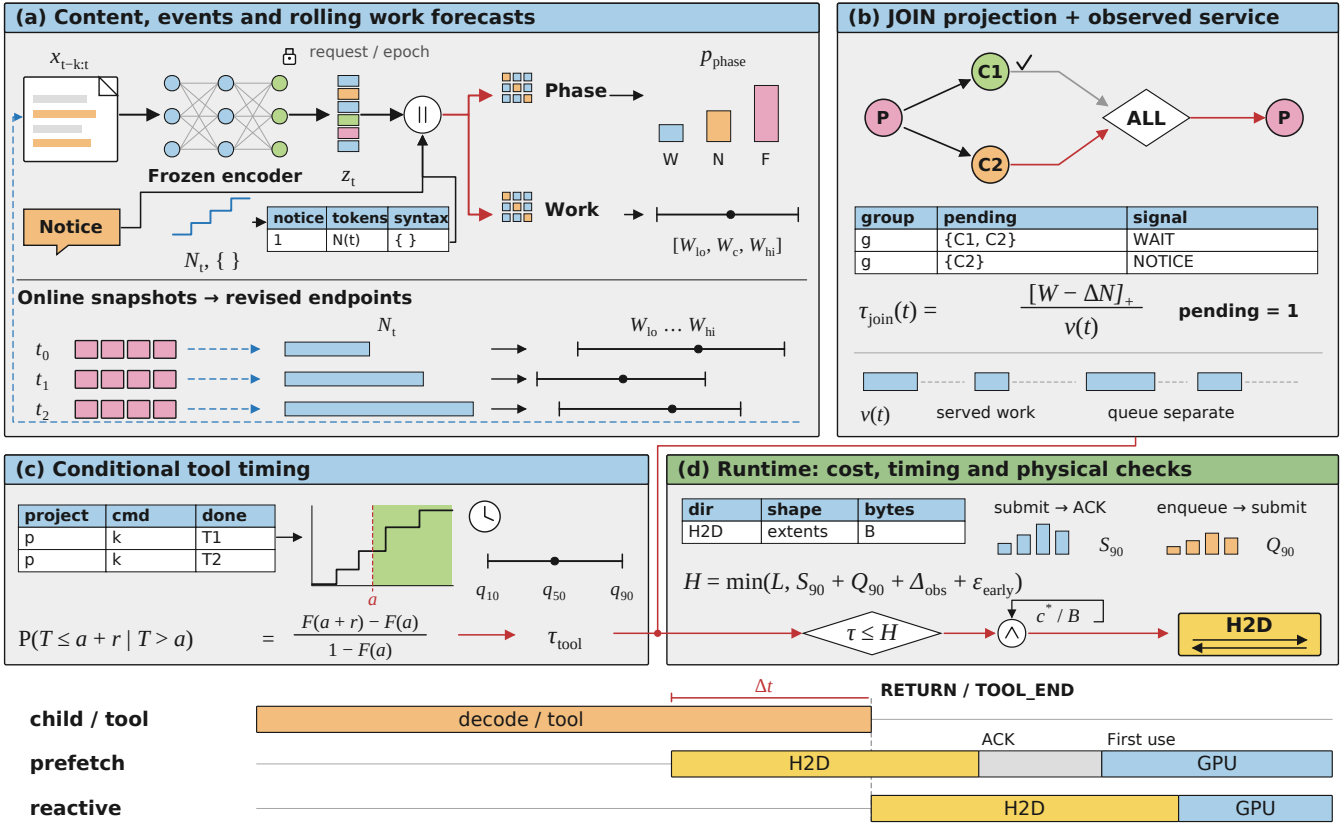
恢复感知的原生准入



Parent　Child 1　Child 2　Runtime 动作　ACK / service / first reuse

From Completion Signals to Near-term Demand

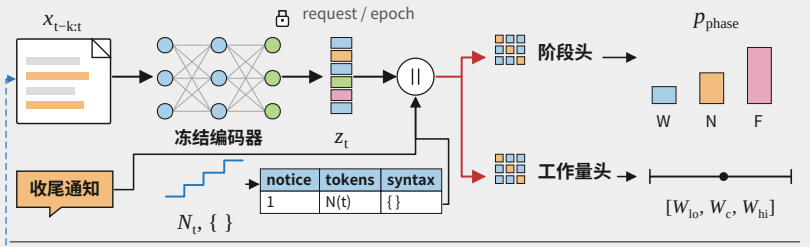
Delivered content + service progress → separate phase / work heads → runtime timing



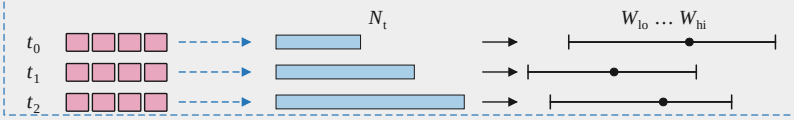
事件与内容驱动的执行需求预测

已送达正文 + 服务进度 → 阶段 / 工作量双头 → Runtime 决定启动时机

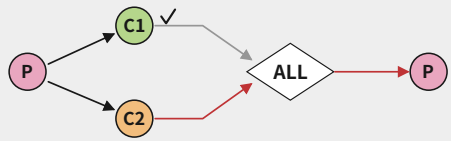
(a) 内容、事件与滚动剩余工作预测



在线快照 → 滚动修正工作端点

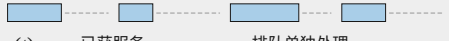


(b) JOIN 投影与已观测服务进度



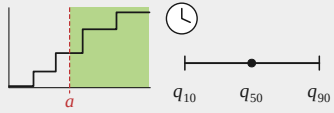
group	pending	signal
g	{C1, C2}	WAIT
g	{C2}	NOTICE

$$\tau_{join}(t) = \frac{[W - \Delta N]_+}{v(t)}$$ pending = 1



(c) 存活条件下的工具时机

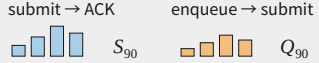
project	cmd	done
p	k	T1
p	k	T2



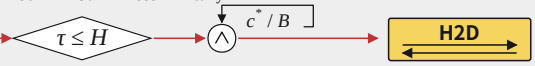
$$P(T \le a + r \mid T > a) = \frac{F(a+r) - F(a)}{1 - F(a)} \rightarrow \tau_{tool}$$

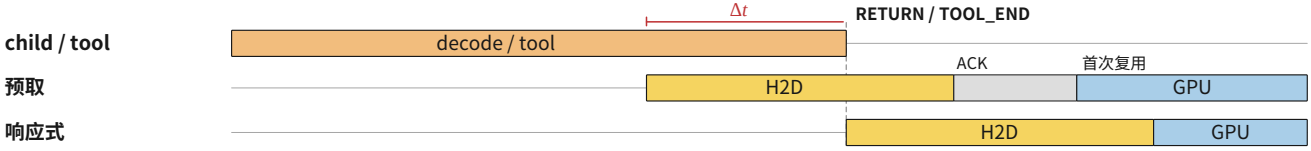
(d) Runtime：成本、时机与物理校验

dir	shape	bytes
H2D	extents	B



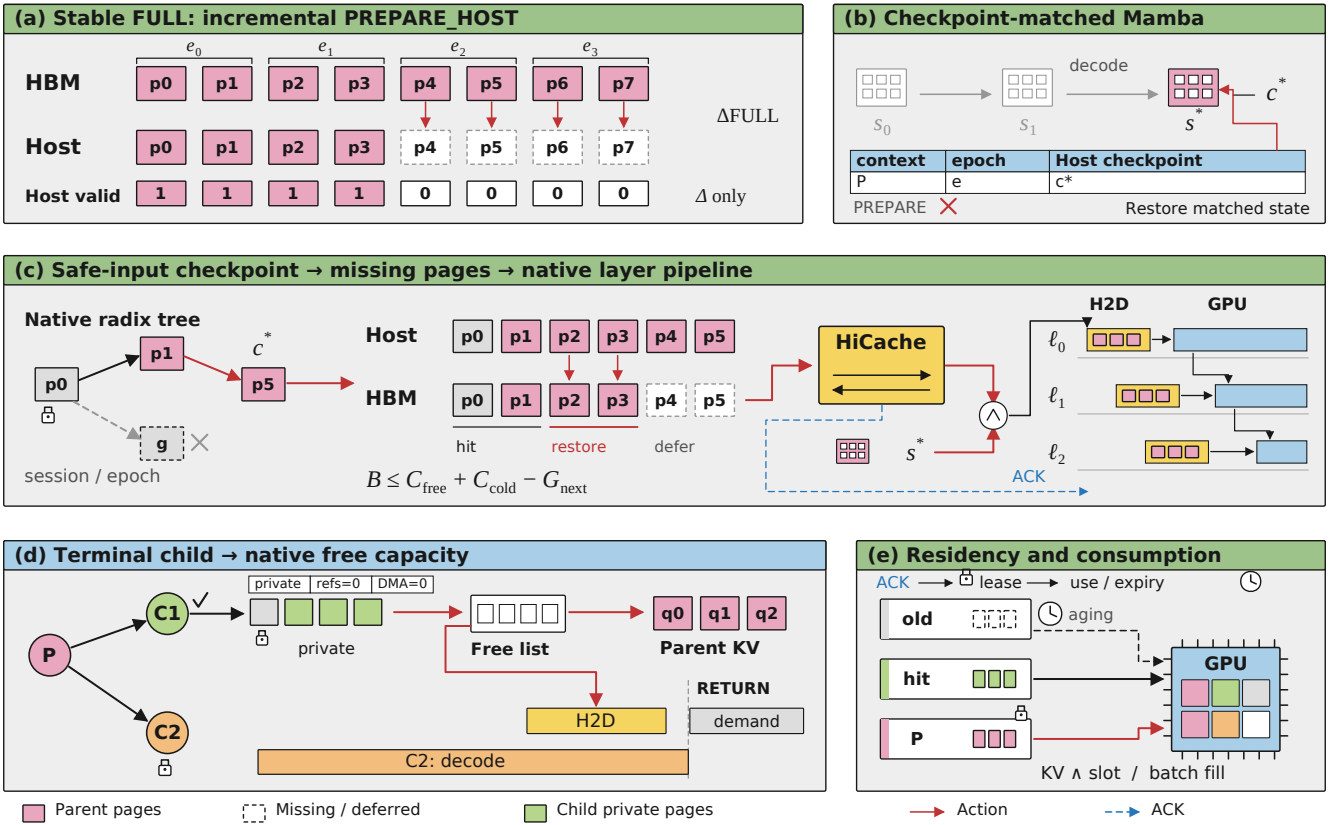
$$H = \min(L, S_{90} + Q_{90} + \Delta_{obs} + \varepsilon_{early})$$





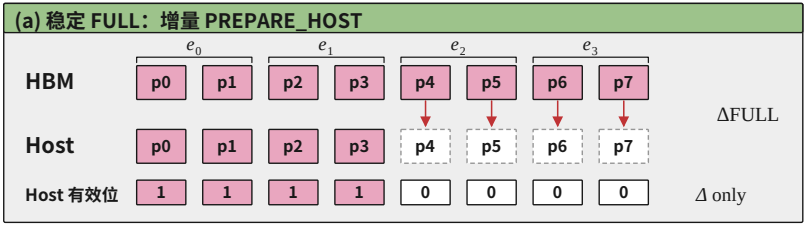
KV Lifecycle and Restore-aware Admission

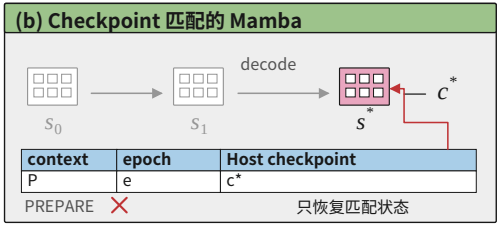
Stable FULL extents · checkpoint state · missing pages · child-to-parent capacity

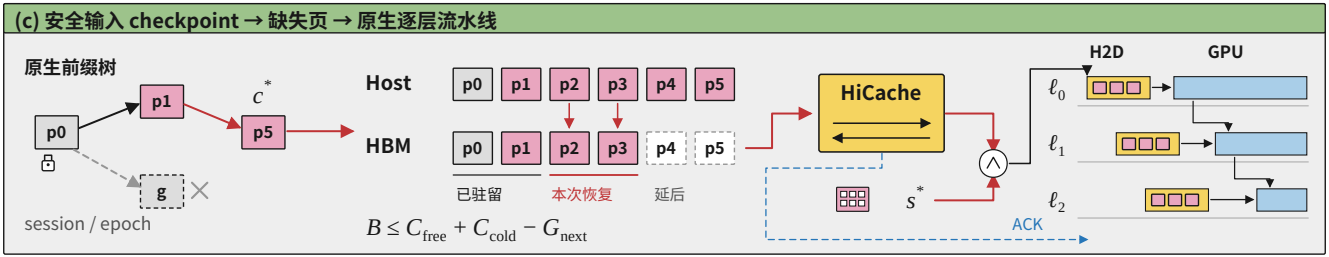


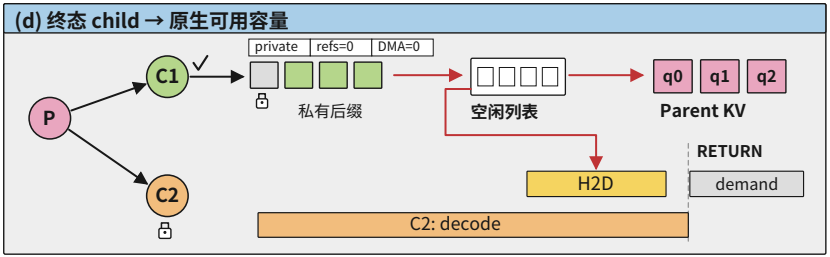
工作流感知的 KV 生命周期与恢复感知准入

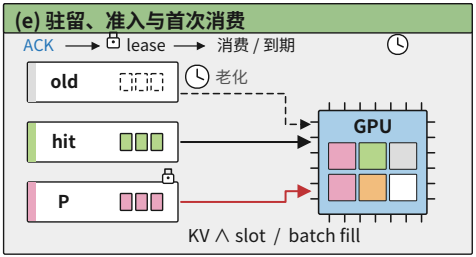
稳定 FULL 前缀 · checkpoint 状态 · 缺失页恢复 · child → parent 容量交接











Parent 页　缺失 / 延后　Child 私有页　控制动作　ACK